

Project 1: David Copperfield Word Network

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Introduction

This project analyzes a word co-occurrence network from Charles Dickens' "David Copperfield". The network has 113 nodes (common adjectives and nouns) and 425 edges connecting words that appear next to each other in the text.

Network Properties

Basic Statistics

- Nodes: 113
- Edges: 425
- Diameter: 5
- Average degree: 7.52
- Connected: No (2 components)

Degree Distribution

The network shows a hub structure with a few highly connected words and many words with low connectivity. Most nodes have 1-10 connections, but some hubs have 40+ connections.

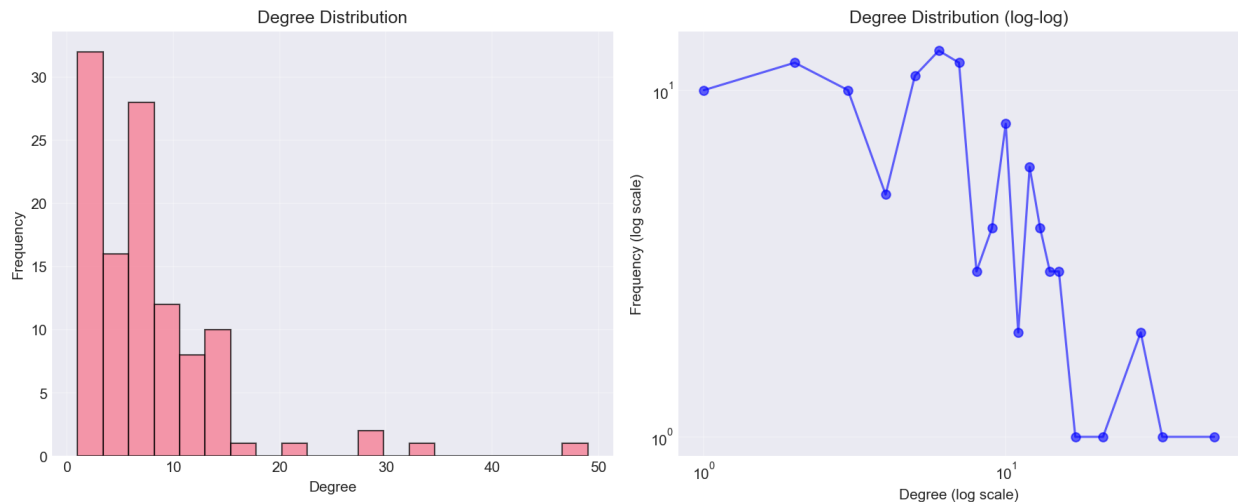


Figure 1: Degree Distribution

Centrality Analysis

Node with Highest Betweenness Centrality

- **Word:** “little”
- **Betweenness:** 0.240316
- **Degree:** 49
- **PageRank:** 0.050438

The word “little” acts as a bridge connecting different parts of the network. It’s a versatile modifier that appears with many different nouns throughout the novel.

Top 5 Nodes by Degree

1. little (49)
2. old (33)
3. other (28)
4. good (28)
5. same (21)

These are all common adjectives that can modify many different nouns, explaining their high connectivity.

Community Detection

Using the Louvain algorithm, I found **8 communities** in the network.

Community 0 (15 nodes): General Descriptors

- **Words:** best, course, family, house, light, love, manner, natural, other, pleasant, side, time, way, whole, world
- **Theme:** Setting descriptions and general qualities

Community 1 (17 nodes): Characters and Emotions

- **Words:** bad, better, boy, child, dear, friend, full, glad, heart, late, mother, poor, pretty, state, true, woman, young
- **Theme:** People and emotional states

Community 2 (23 nodes): Time and Character Descriptions

- **Words:** agreeable, air, alone, aunt, certain, common, day, early, evening, first, greater, happy, life, lost, man, miserable, moment, money, morning, name, old, person, year
- **Theme:** Temporal references mixed with character descriptors

Community 5 (20 nodes): Physical Descriptions

- **Words:** bed, bright, door, eye, face, fire, hand, head, hope, kind, large, little, new, open, red, round, small, something, strange, white
- **Theme:** Body parts, sizes, and colors

Community 6 (16 nodes): Abstract Concepts

- **Words:** beautiful, black, dark, good, great, hard, letter, long, master, mind, night, part, perfect, thing, word, work
- **Theme:** Mix of physical and abstract descriptors

David Copperfield Word Network - Communities

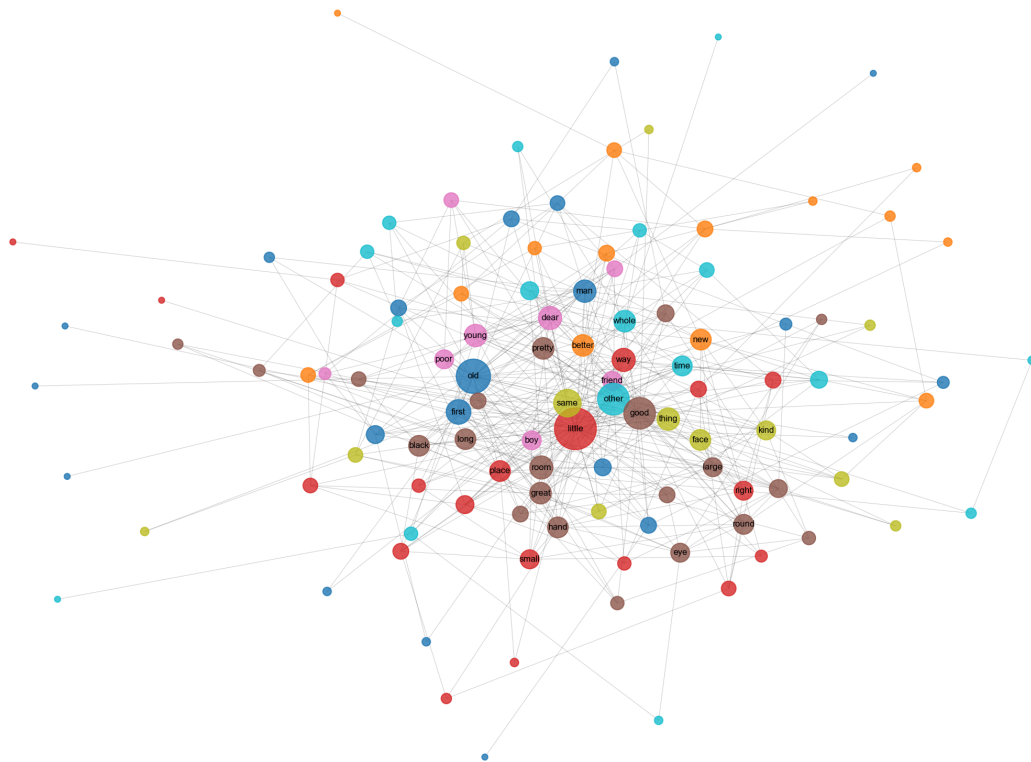


Figure 2: Network Communities

Interesting Observations

Why are different words grouped together?

Communities form based on co-occurrence (words appearing together), not meaning. Some examples:

- “money” and “morning” are in the same community because they appear in similar contexts in the novel
- “fire” and “face” cluster together because Dickens often describes faces lit by firelight
- “hand” and “hope” appear together in phrases about gestures or hopeful actions

This reveals more about Dickens’ writing style and common phrase patterns than about word meanings.

Ego Network Analysis

Selected Word: “little”

I chose this word because it has the highest degree, betweenness, and PageRank in the network. It’s the most central word.

Properties

- Degree: 49
- PageRank: 0.050438
- Community: 5

Ego Network Structure

- **Depth 0:** 1 node (little itself)
- **Depth 1:** 49 direct neighbors
- **Depth 2:** 61 second-degree neighbors
- **Depth 3+:** 2 remaining nodes

The ego network at depth 2 covers 111/113 nodes (98%), showing how central “little” is.

Direct Neighbors

Direct neighbors include: face, hand, room, door, way, time, boy, man, woman, etc.

This shows “little” is used to modify many different types of words - people, places, body parts, and abstract concepts.

Gephi Visualization

Layout

- **Algorithm:** ForceAtlas 2
- **Settings:** Moderate gravity, overlap prevention enabled
- **Rationale:** Good for showing community structure while keeping nodes readable

Node Sizing

- **Attribute:** By degree
- **Effect:** Larger nodes = more connections
- **Benefit:** Makes hubs immediately visible

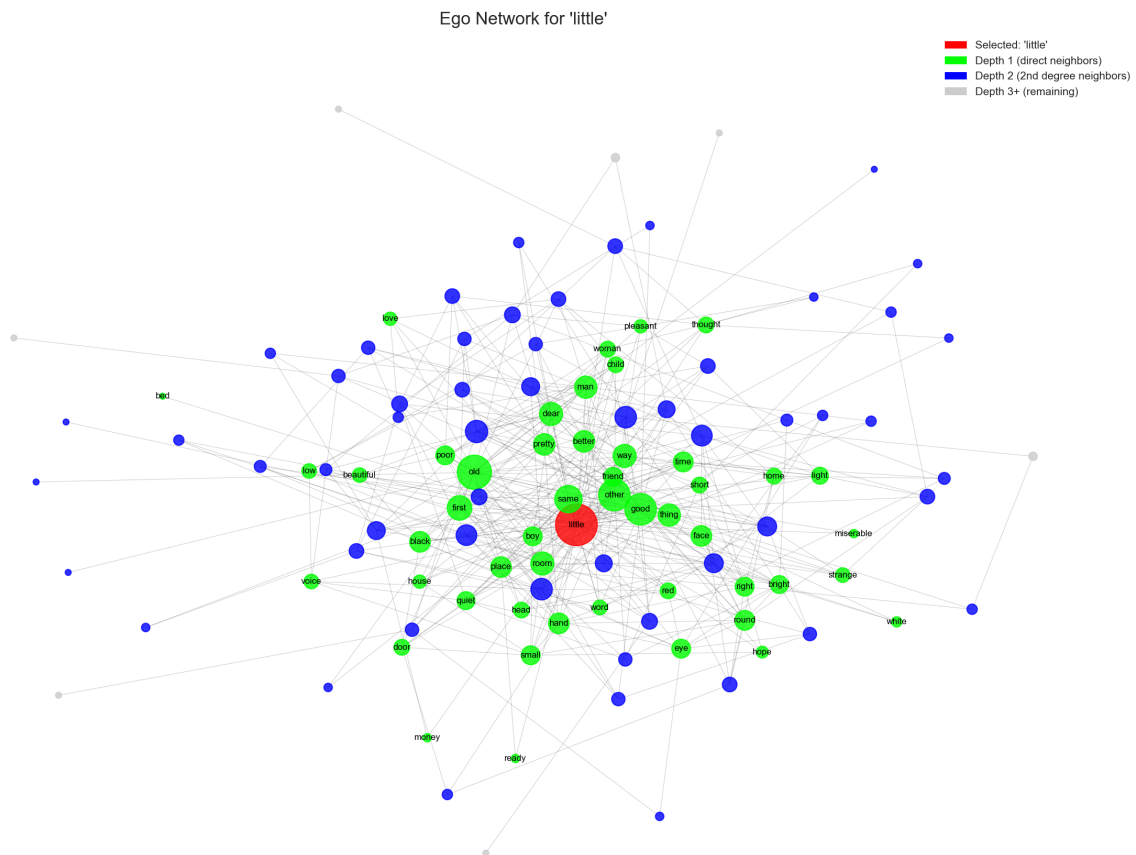


Figure 3: Ego Network

Node Coloring

- **Attribute:** By community (8 colors)
- **Scheme:** Each color = one community from Louvain algorithm
- **Rationale:** High contrast for easy distinction

Edges

- **Style:** Light gray, thin
- **Rationale:** Subtle to avoid clutter

Screenshots

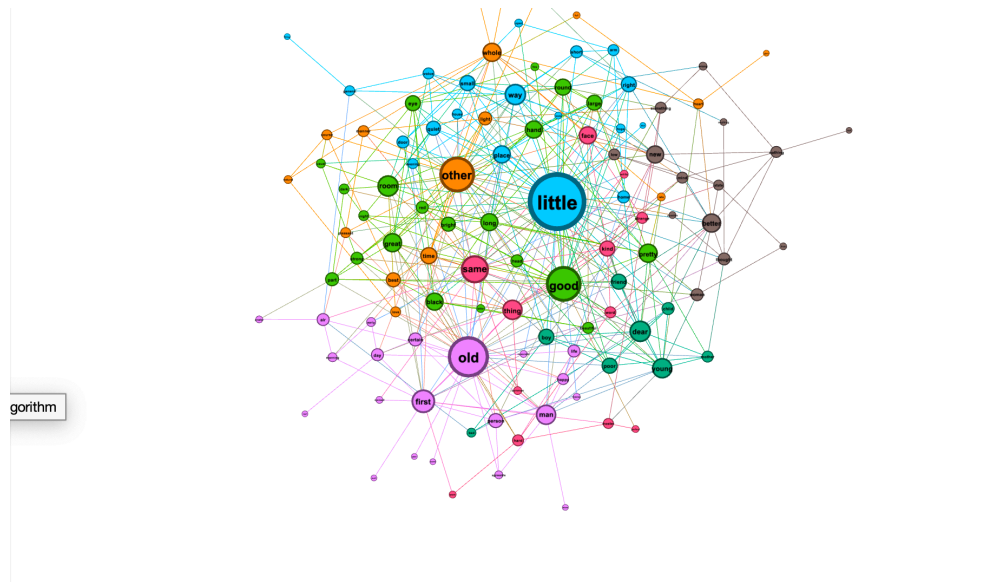


Figure 4: Full network showing community structure

Discussion

Key Findings

1. **Hub structure:** A few central adjectives (little, old, good) connect most of the network
2. **Communities reflect usage:** Words cluster by how Dickens uses them together, not by meaning
3. **“Little” dominates:** Most central by all metrics, covering 98% of network at 2 hops
4. **Victorian style visible:** Physical descriptions (face, hand, fire) and domestic settings (house, room) are prominent

Limitations

- Only common words included (rare words excluded)
- Adjacency doesn't always mean meaningful relationship
- No directionality (don't know which word modifies which)
- Context and sentence structure lost

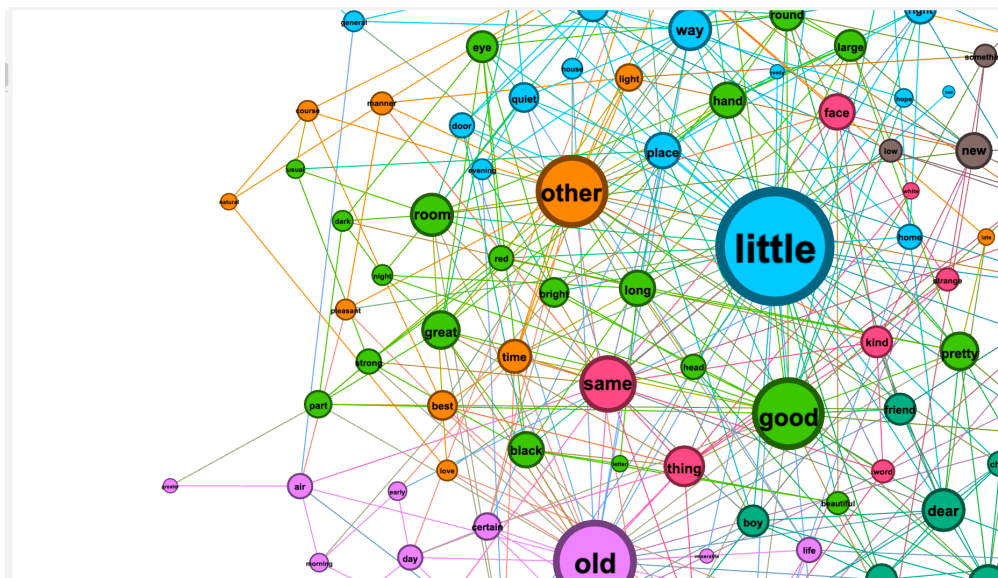


Figure 5: Clean layout with no node overlaps

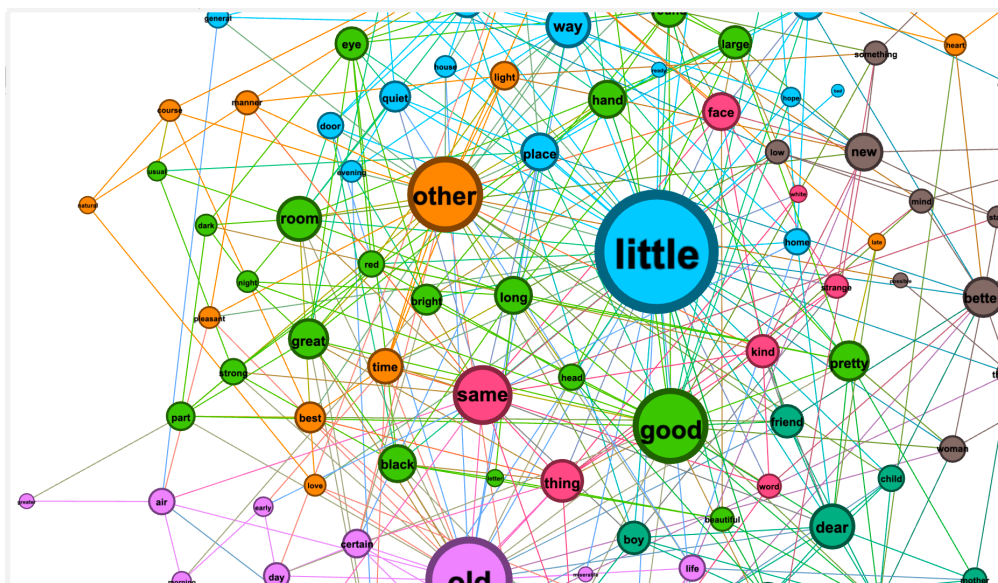


Figure 6: Clear size differences showing degree centrality

Conclusion

Network analysis reveals patterns in Dickens' writing that aren't obvious from reading. The dominance of "little" and other versatile adjectives, the clustering of physical and emotional descriptors, and the integration of character descriptions with temporal references all reflect Victorian literary style.

The Gephi visualization and Sigma.js export make these patterns easy to explore interactively.

Technical Details

Tools

- Python 3 + NetworkX, Pandas, Matplotlib, Seaborn
- Louvain algorithm for community detection
- Gephi 0.10 for visualization
- Jupyter Notebook for analysis

Files

- `analysis.ipynb` - Full analysis with code
- `david_copperfield_network.gexf` - Network export
- `david_copperfield_network.gephi` - Gephi project
- `network/` - Interactive Sigma.js visualization

Data

- `nodes.csv` (113 nodes)
- `edges.csv` (425 edges)
- Source: Charles Dickens, "David Copperfield" (1850)